

Listen, Predict, and React

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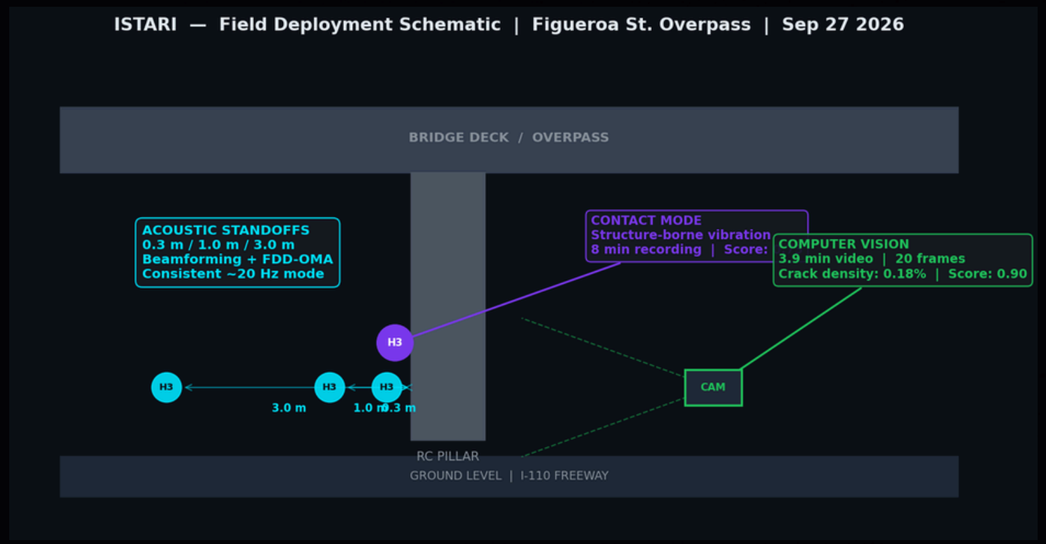
Problem & Customer

Prompt D: How can we detect infrastructure damage and failures before they become expensive, and rapidly prioritize recovery when they occur?

Problem: Utilities, telecommunications companies, transportation networks, insurers, and industrial operators spend over **$50B annually** inspecting and maintaining physical assets. The current workflow is:

- **MANUAL & REACTIVE**
- **SURFACE LIMITED**
- **COSTLY & SLOW**

CORE FAILURE: Every current inspection system is blind to what matters most -- what is happening inside the structure. A beam can be structurally compromised for months before any camera can detect it.



Target customers:

- Transportation agencies and DOTs
- Electric utilities and grid operators
- Industrial and manufacturing facilities
- Government and defense installations
- Nuclear facility operators

Customer Insights

Interviews:

- Jayesh Patel: 35 years Architect GB98 Certified Contractor
- Parin Bhakta: 15 years experienced Project Manager/Superintendent
- Senior Officer, Indian Air Force HQ: validated acoustic intelligence for predictive structural maintenance

Market Anaylsis:

- Global SHM market: $3.9B in 2026, projected $8.6B by 2035 at 9.1% CAGR
- 623,000 US bridges: 49.1% in fair condition and aging; 6.8% structurally deficient (ASCE 2025)
- FHWA mandates biennial inspections at $4,500-$10,000 each -- 600K+ inspections due every 2 years

Information / Advice Learned:

- Target: DOTs, utilities, industrial and defense operators -- all face mandated inspection cycles
- Inspector deficit: NBI certification takes 18-24 months; sector short 500K workers by 2026.
- Key Bridge collapse (2024): $1.7B rebuild + $15M/day loss-failure dwarfs inspection cost.
- FHWA biennial mandate = recurring contract pipeline across all 50 states.

Needs / Possible Additions:

- Real-time continuous monitoring dashboard
- Multi-asset portfolio triage capability
- Drone-mountable sensor configuration
- Post-disaster rapid deployment mode

Motivations to Buy / Pricing:

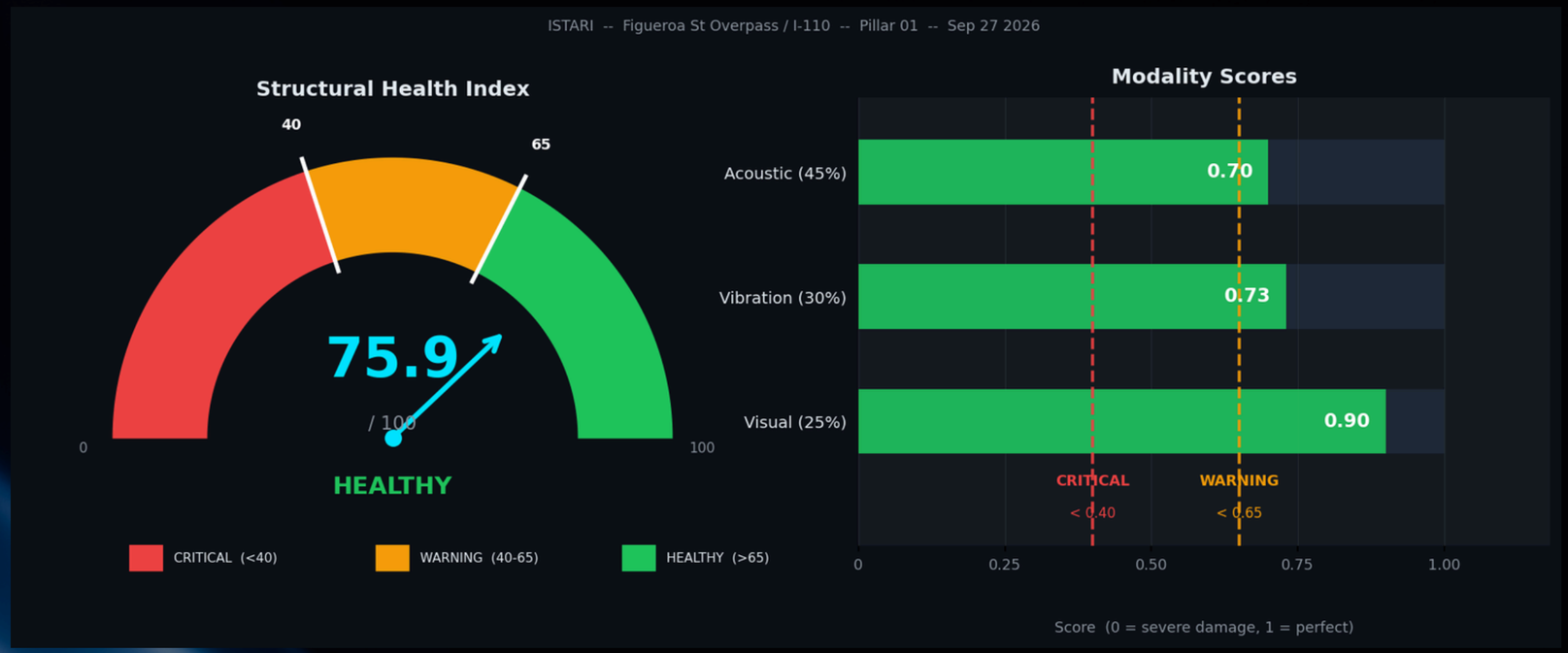
- Cost of failure far exceeds inspection cost
- Liability reduction and regulatory compliance
- Reduces crew deployments and site visit costs
- Per-project from $35K; portfolio from $150K; enterprise custom

Solution MVP

ISTARI is a three-modality sensing platform that detects infrastructure damage before it becomes visible -- and before it becomes catastrophic. Our system uses acoustic sensing, vibration analysis, and computer vision in a fused pipeline that produces a single, actionable health score for any structural element.

Addresses the Problem By:

- Reactive: continuous sensing predicts failure before it happens, not after.
- Blind: acoustic/vibration catch internal damage before it surfaces; vision covers the rest.
- Costly: one health score replaces manual crew visits; scales to triage thousands of assets after a disaster.
- Disaster Response Gap: instant prioritization of which assets need inspection first, cutting unnecessary downtime.



Value Proposition

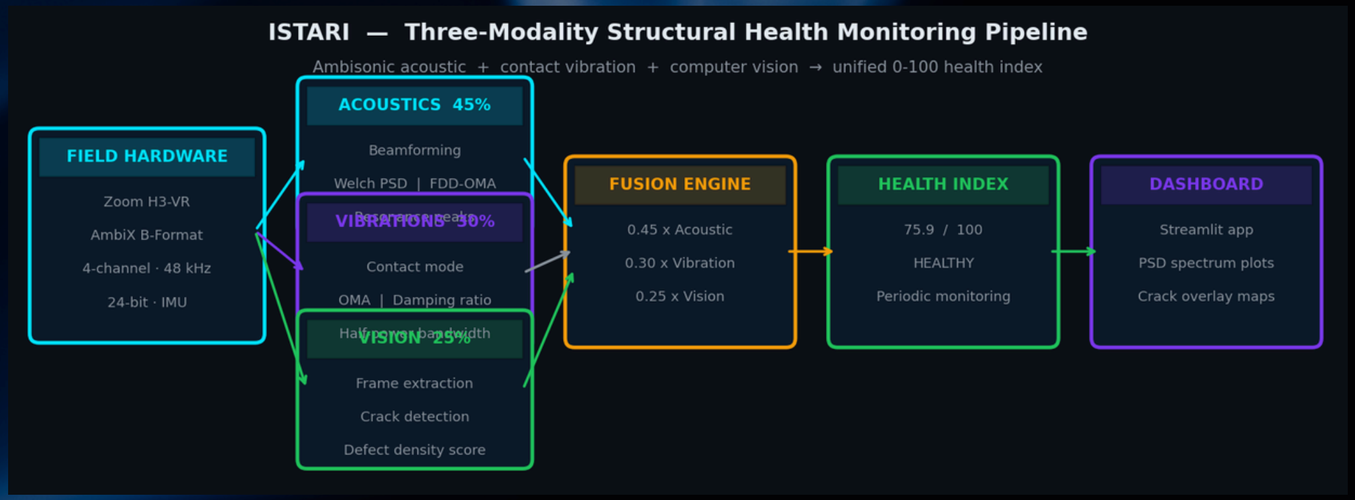
Core idea: Structural failure happens catastrophically and without warning, because the damage that matters most is invisible, hidden inside the structure, undetectable by any camera or manual inspection. ISTARI is different because it senses that internal damage directly, before it becomes a visible crack or a catastrophic failure.

The problem with current methods:

- Reactive: inspections follow fixed schedules, not actual condition; failures are found after they happen, not predicted before.
- Blind: visual inspection only catches surface damage; by the time a crack shows, internal compromise may be 40–60% advanced.
- Costly: inspecting remote/elevated structures costs thousands per visit and doesn't scale for post-disaster triage of thousands of assets.

Why it matters most in a crisis:

- Disasters (wildfire, earthquake, flood) create two crises at once: mass asset assessment + compounding downtime costs.
- LA wildfires example: no way to prioritize which power towers to inspect first; days of unnecessary outage.
- ISTARI: three sensing modalities; one unified health score, more reliable than any single method
- Result: prioritize and act before failure, not after.



Business Model

Who pays?

- Government / state transportation agencies
- Utilities
- Infrastructure owners
- Large contractors and developers
- Industrial operators

Why do they pay?

- Reduce manual inspection labor
- Reduce unnecessary truck rolls / site visits
- Identify high-priority assets faster
- Reduce downtime
- Improve documentation and risk management

Initial Revenue Model

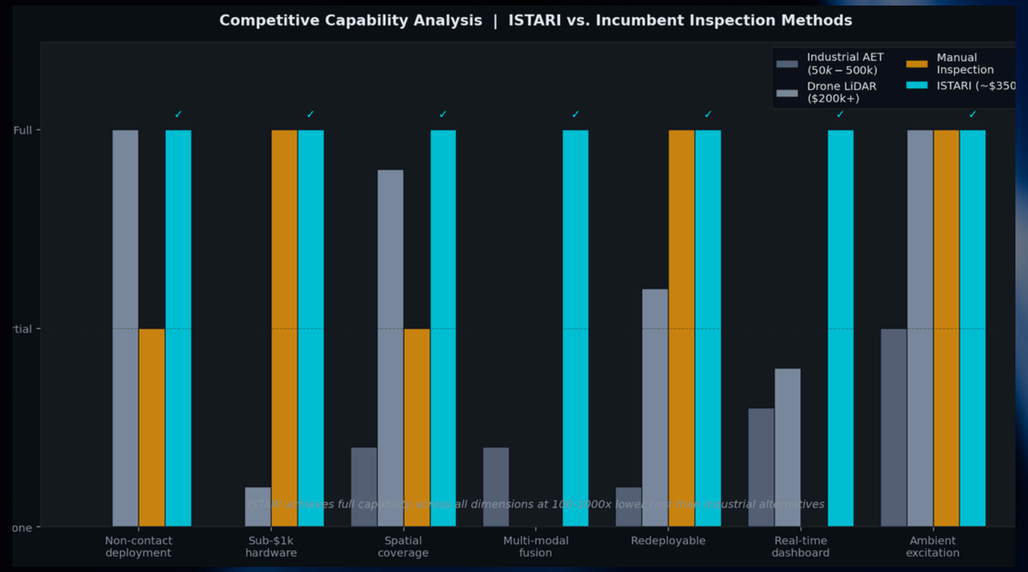
Inspection-as-a-Service

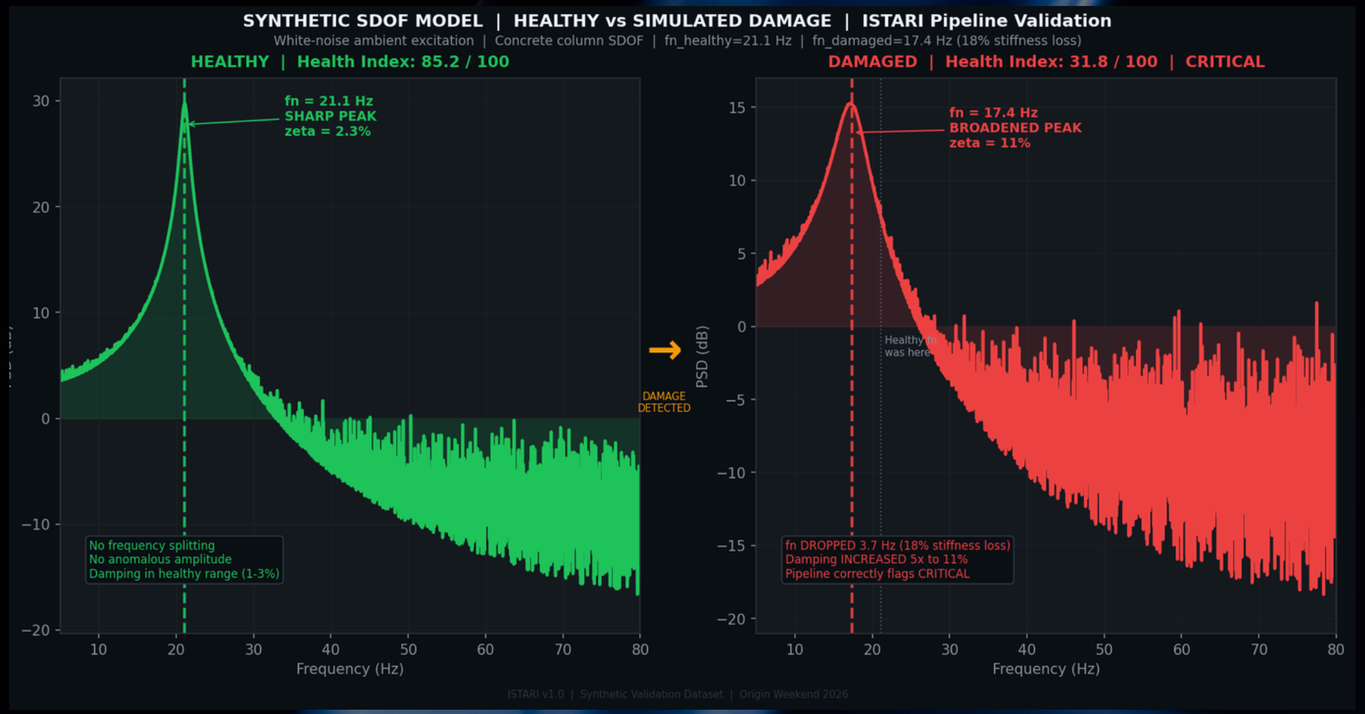
- Customer pays per asset, project, or inspection campaign
- ISTARI provides sensors, data collection, analysis, and report

B2B SaaS + Hardware

- Annual platform subscription
- Per-asset monitoring fee
- Hardware / sensor package
- Premium disaster-response inspections

Initial hypothesis: $35K –$75K per inspection campaign; $150K –$400K for multi-asset portfolio





Next Steps

Validate:

- Test whether acoustic/vibration anomalies correlate with known structural changes.
- Compare ISTARI results against expert inspection.
- Measure false positives / false negatives.
- Conduct more interviews with utilities, DOTs, and infrastructure owners.

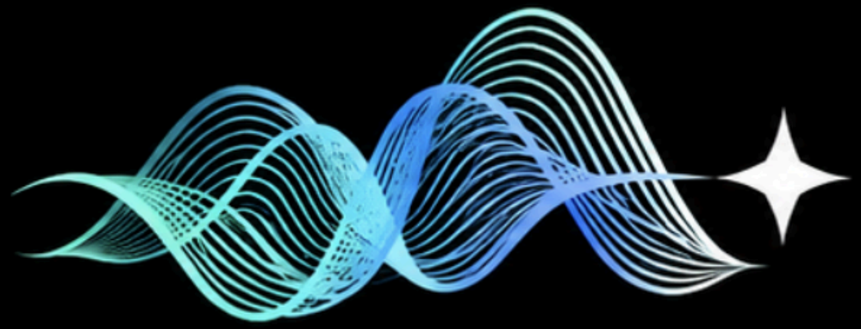
Build:

- Improve acoustic and vibration signal processing.
- Improve computer-vision defect detection.
- Automate sensor fusion and health scoring.
- Build before/after asset comparison.
- Develop the operator dashboard.

Scale:

- Test across different infrastructure types.
- Establish baseline measurements for each asset.
- Integrate repeated inspections over time.
- Pilot with an infrastructure owner or government agency.

Thank You



ISTARI

INFRASTRUCTURE INTELLIGENCE